Ole A. Ellingsberg Helmers, 08.04.2002
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Education

- Norwegian University of Science and Technology (NTNU), Trondheim
  - Cybernetics and Robotics - Master's Program (5-year), 2022 - 2027
- Setesdal Folk High School, Climbing and Outdoor Life, 2021-2022
- Elvebakken High School, Oslo, 2018-2021

Work Experience and Technical Experience

- WaterLinked project engineer - team leader - in R&D summer 2026
  - Proposed and lead an initiative to demonstrate target tracking on the Sonar 3D-15.
  - Proposed the implemented design of the target tracking system to perform real time tracking of multiple ROVs in a cluttered environment.
  - Created a proof of concept of velocity and position estimation of an ROV using a mounted Sonar 3D-15 using the tracker to track multiple stationary targets using Python.
- WaterLinked project engineer in R&D summer 2025
  - SLAM using 3D-sonar and DVL on various industrial underwater ROVs. My contribution was that I designed a 3D-SLAM front-end, and parts of the back-end
  - Finished the ROS-2 driver for the Sonar 3D-15
  - Technical writing of 3D-sonar documentation
- Technical Position in Vortex NTNU Software, Perception Team Member, 2024 to 2026
  - Year 1: Developed a ROV-vision pipeline for detecting the 3D-position and rotation of the gripping point of an underwater valve. Using YOLOv8 and geometrical filtering of a depth-image from a stereo camera using classic robot vision techniques like RANSAC and canny edge detection. I mainly used C++ and OpenCV.
  - Year 2: Developed a 2D-SLAM back-end using iSAM2 for a multi-beam 2D sonar system in ROS 2, validated via Stonefish simulation.
- Omega Workshop, Board Member, from 2022 to 2025
- Omega Workshop IT Committee: 2022 to 2025

Academical interests and experience

- The fields of study that interest me the most:
  - Perception including computer and robotic vision and sensor fusion
  - Robust/safe navigation and control
  - Modeling and simulation
  - Nonlinear systems
  - Reinforcement-learning
- Some experience from solo projects
  - 5 legged robot walking gait using reinforcement learning: https://github.com/oahelmer/FiveLeggedRobotRLwalk
  - 2D-boat simulator controlled by reinforcement learning and LOS guidance: https://github.com/oahelmer/2d-boatsim
  - Quadcopter simulation using MPC control: https://github.com/oahelmer/Quadcopter-Simulation

Languages

- Norwegian (native)
- English (fluent)

References available upon request